

Data-quality & EDA insight memo

To Head of Operations, Merchandiser, Finance lead - NorthBay Living
From FORESIGHT data scientist (Zidio Development)
Data Daily sales 1 Jan 2024 - 31 Dec 2025 • 50 SKUs • monthly stock snapshots to 01 Dec 2025

In one minute

- 1. **Demand is seasonal and steady.** Every year peaks in Mar (+23% vs an average month) and bottoms out in Oct (-21%). Year-on-year volume is flat (-0.1%).
- 2. **Stock is not where demand is.** 7 SKUs are about to run short while 3 SKUs hold more than 90 days of stock (₹48.7 L at cost).
- 3. **Promotions and weekends are big, predictable levers.** Promotion weekends sell 24% more than normal weekends, which already sell 15% more than weekdays.
- 4. **16 SKUs are priced below their recorded cost** (17% of revenue). Finance should confirm the cost data.
- 5. **The extracts need fixes at source.** There are 150 stock-only SKUs, launch dates that come after first sales, mismatched subcategories, and stock values that don't match cost. All are handled in code and listed in section 1.

1 Data quality: what we found and how we handled it

All four extracts are ingested and cleaned by `src/pipeline.py`. No step is done by hand, and a re-run gives the same result. The pipeline checked for the missing values, duplicates and inconsistent labels the brief warned about. These checks found nothing: Duplicate SKU rows; Unparseable dates; Missing numeric values; Duplicate (date, SKU) rows; Negative units; Revenue != units x price; SKUs not in `sku_master`; Duplicate dates; Sales promo_flag vs calendar promotion days; Sales dates missing from calendar; Duplicate (date, SKU) rows; Missing or negative stock values. The problems it did find:

Table	Issue and why it matters	Rows	What we did
sku_master	Unlabelled column 'wfr' The brief's dictionary expects unit_cost; the extract ships it under a cryptic name.	50	Renamed to unit_cost (identity price - wfr = margin holds for 50/50 rows)
sku_master	Unit cost above selling price (negative margin) Could be a real pricing problem or a cost-data error - the client must confirm; we do not change prices (price optimisation is out of scope).	16	Kept as-is and flagged (negative_margin=True)
sku_master	Category/subcategory labels inconsistent Pairs like Kitchen > Cushion or Storage > Sofa cannot both be right and we cannot tell which label is wrong, so we group by category only.	30	Kept; category used as the grouping level, subcategory flagged as unreliable
sales_daily	Sales dated before the SKU's launch_date Pre-launch sales volumes look identical to post-launch ones, so the sales are real and the launch dates are wrong. Effective launch = first sale date.	3,393	Kept the sales; launch_date treated as unreliable (21 SKUs affected)
calendar	Provided 'week' column is not a standard week number Week 1 of 2025 has 5 days and week 1 of 2024 has 7, so it is not ISO-consistent.	731	Not used; weeks rebuilt as Monday-Sunday from the date
inventory_snapshots	SKUs with stock but no master data or sales No sales history or cost -> cannot forecast or value them. Client should confirm whether these are discontinued items or a separate catalogue.	3,600	Dropped (150 SKUs: SKU051-SKU200)
inventory_snapshots	Inventory_Value does not equal stock x unit cost The supplied value swings from 0.05x to 15x stock-at-cost, so it	1,152	Recomputed as on_hand_units x unit_cost

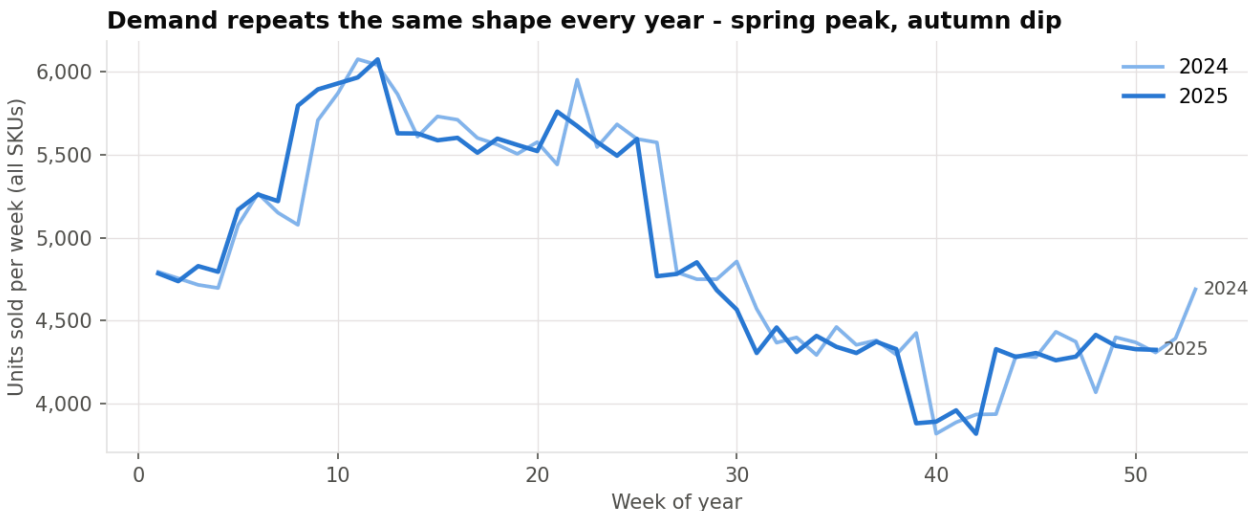
cannot be trusted for rupee impact.

inventory_snapshots	Snapshots are monthly and do not roll forward from sales Month-to-month stock jumps are unrelated to sales, so projecting stock from sales would invent numbers. We score risk as of the latest snapshot date instead.	24	Latest snapshot used as the current stock position; no roll-forward
sales_daily	Incomplete weeks at the edge of the data A 3-day week would look like a demand collapse to the model.	150	Dropped from the weekly panel (week(s) starting 2025-12-29)

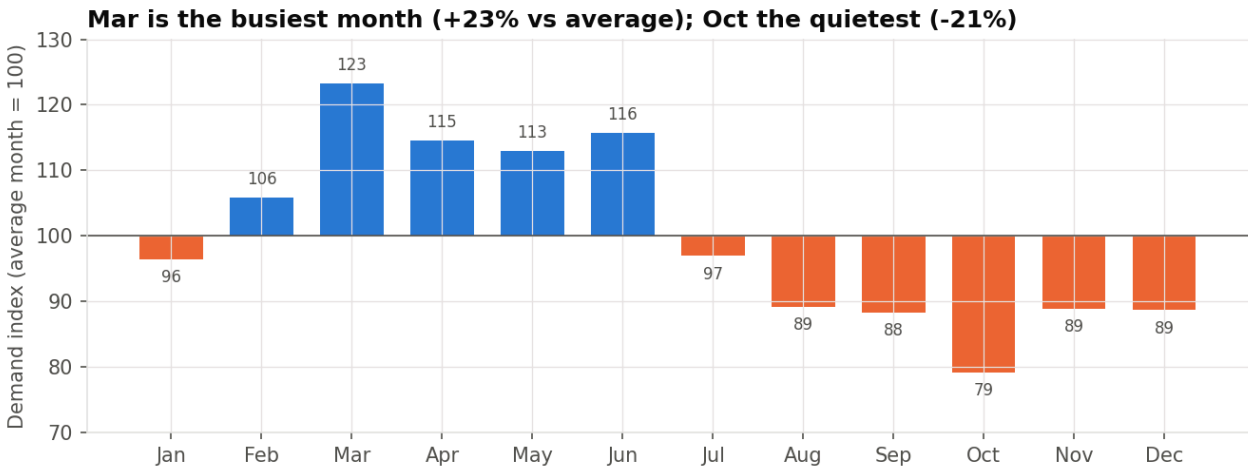
Full log with rationale: outputs/data_quality_log.csv. Two raw files arrived as Excel despite a .csv name. The loader reads them by content.

2 Demand patterns

Seasonality and trend

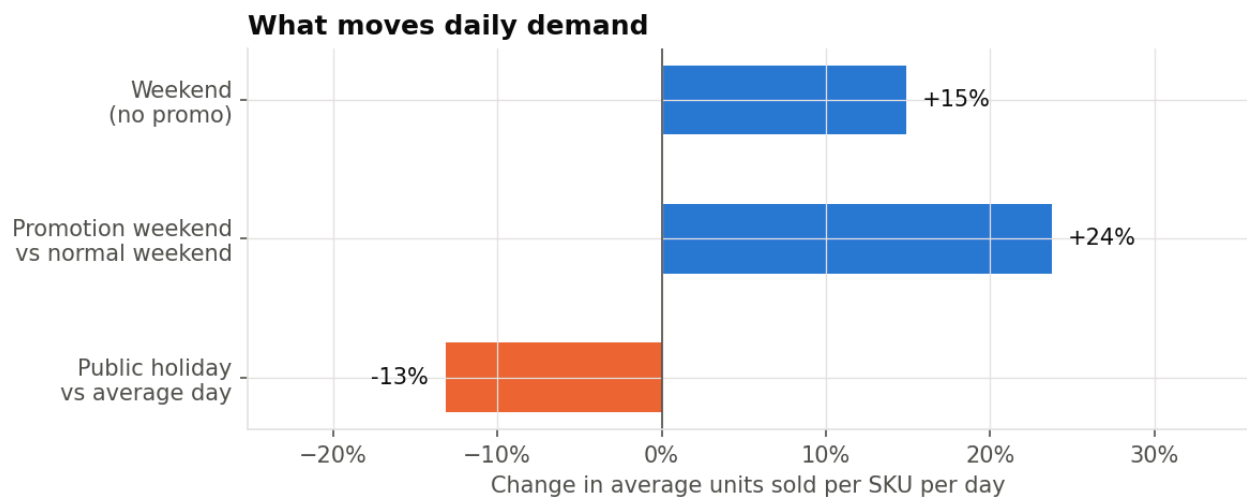


Total units per week, all 50 SKUs. The 2025 line sits almost on top of 2024. Weekly totals range from 3,817 to 6,076 units.



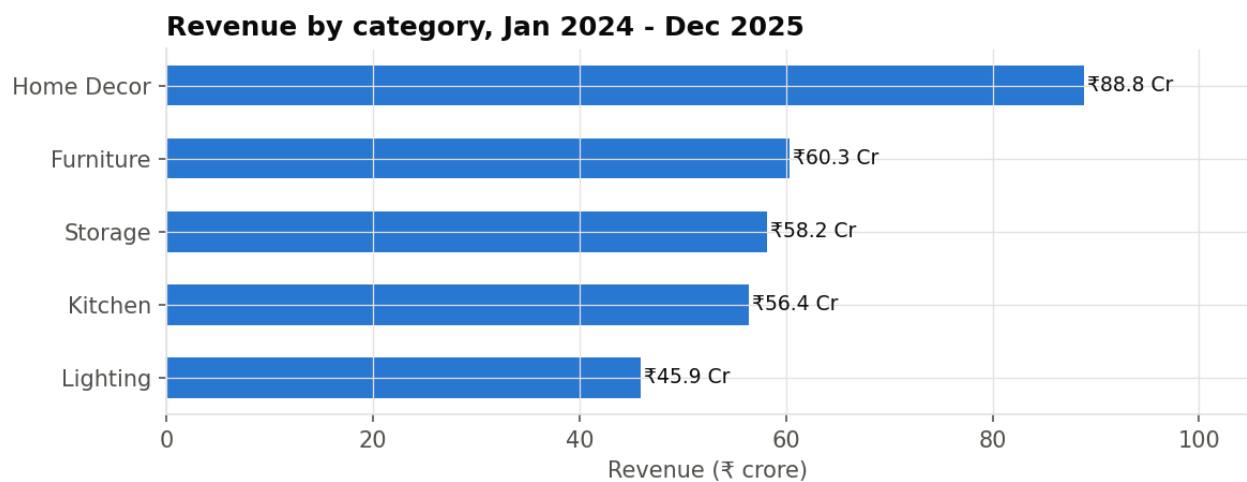
Average daily units per SKU in each month, indexed to 100. Spring (Feb-Jun) runs above average and Aug-Dec below.

What moves demand day to day

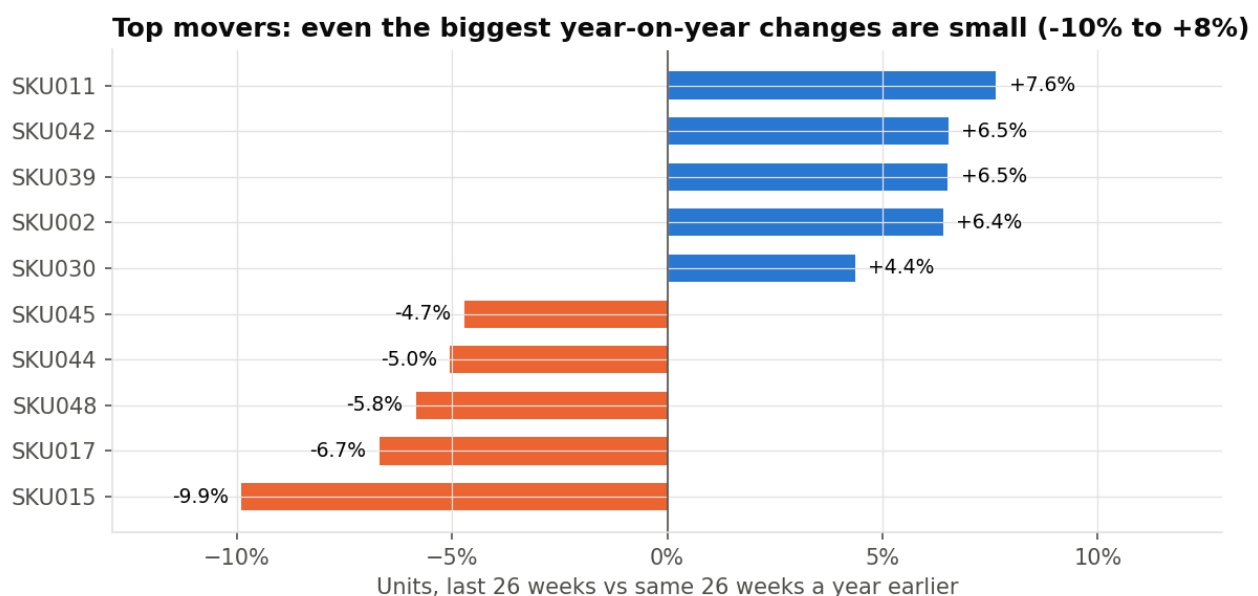


Promotions run store-wide on weekends in March, June, September and November. They are compared with normal weekends so the weekend effect isn't counted twice. Holidays (Republic Day, Independence Day, Diwali, Christmas) are compared with an average day.

Categories, best-sellers and top movers

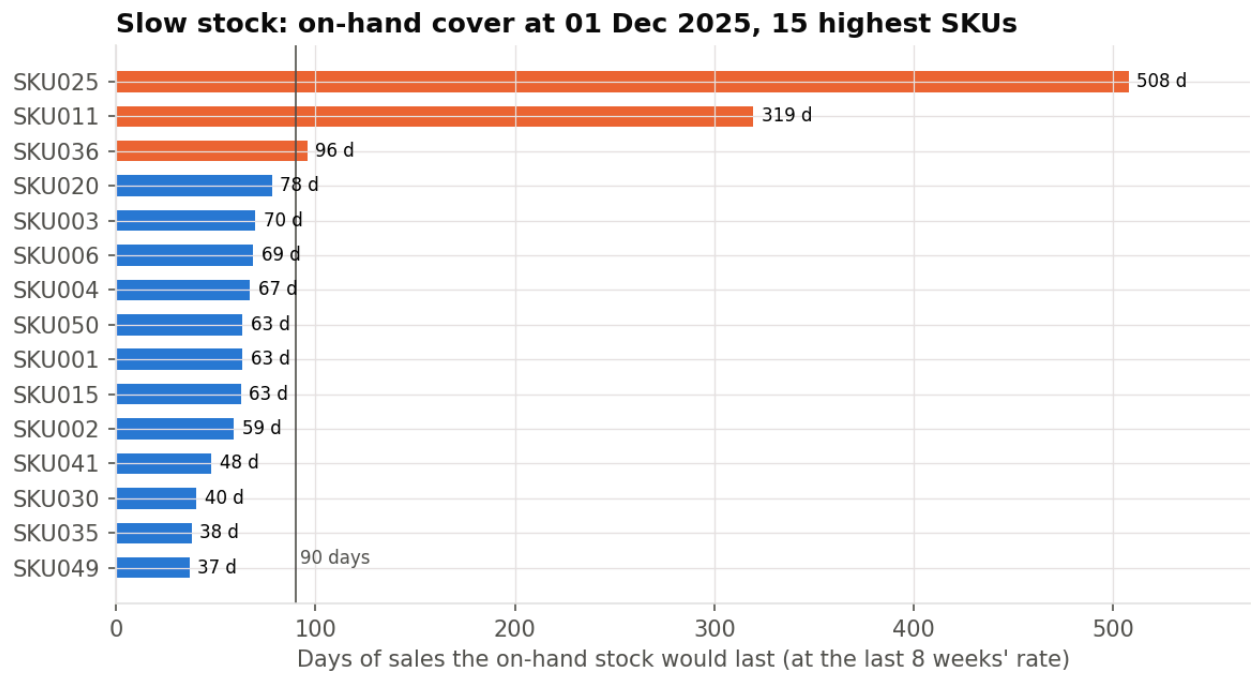


Home Decor is the largest category (₹88.8 Cr over two years). Revenue is concentrated: the top 10 SKUs bring in 48% of revenue, and 24 SKUs make up 80%.



Biggest risers: SKU011 +7.6%, SKU042 +6.5%, SKU039 +6.5%. Biggest fallers: SKU015 -9.9%, SKU017 -6.7%, SKU048 -5.8%. No SKU moved more than 10%. Price and volume are barely related across SKUs (correlation 0.10).

Dead and slow stock



Median SKU holds 22 days of stock. Over 90 days: SKU025 (508 days), SKU011 (319 days), SKU036 (96 days). Beyond these, 150 SKUs (SKU051-SKU200) hold 39,985 units but have no sales at all. They may be dead stock or a separate catalogue.

3 Business insights

1 • Plan by season, not by last month

Demand repeats the same yearly shape: Mar is 44 index points above Oct, and year-on-year growth is flat. Ordering by "what we sold last month" leaves the team short going into spring and overstocked going into autumn.

So what: build stock from February for the Mar-Jun peak, and let it run down from July.

2 • 7 SKUs need stock now while 3 hold more than 90 days of it

At the 01 Dec snapshot, 7 SKUs are flagged to reorder. 5 of them (SKU012, SKU031, SKU017, SKU040, SKU010) have less stock, on hand and on order, than they will sell before a new order can arrive. At the other end, SKU025, SKU011 hold 508 and 319 days of stock. The problem is stock allocation, not total stock.

So what: use the FORESIGHT reorder and markdown lists each month rather than a blanket reorder.

3 • Promotions are worth planning stock around

A promotion weekend adds 24% on top of an already-strong weekend (+15% vs weekdays). Holidays depress sales (-13%). Promotions fall on the same weekends every year.

So what: stock up best-sellers before the March, June, September and November promotion weekends. The forecast already includes planned promotions.

4 • 16 SKUs lose money on every sale, if the cost data is right

For these SKUs the product master records a unit cost above the selling price. They bring in 17% of revenue. At the recorded costs they would have lost ₹30.38 Cr in gross margin over two years. 3 of them (SKU040, SKU023, SKU010) are on this month's reorder list.

So what: Finance should confirm these costs before those SKUs are reordered. Pricing decisions are outside this engagement.

5 • The stock file needs a clean-up at source

Month-to-month stock levels don't follow from sales and deliveries: only 9% of monthly moves are within 20% of what sales and on-order stock would predict. The stock value column doesn't equal stock × cost, and 150 SKUs appear only in the stock file.

So what: FORESIGHT scores risk from the latest snapshot only and recalculates values from unit cost. A reliable weekly stock export would make the risk flags sharper.

4 What this means for the forecast

Strong yearly seasonality makes "same week last year" a demanding baseline to beat. Weekends, promotions, holidays and recent level are the obvious features to add. The forecast is weekly by SKU, 8 weeks ahead, and is scored on WAPE (average miss as a share of actual demand). It beats that baseline in every backtest window: 8.7% vs 10.5% (see the executive readout and README).